

VSB Event Portal - SQL Database Schema

Relational Database Schema Table Creation Queries (DDL)

The VSB Event Portal utilizes Google Cloud Firestore (NoSQL) for serverless client-side operations. Below is the mapped Relational SQL DDL (Data Definition Language) Schema representation required for structured database system integration and report submissions. All Primary Keys, Foreign Keys, and Cascading Deletes are mapped accurately.

SQL Table Creation Queries

```
-- SQL SCHEMA CREATION SCRIPT FOR VSB EVENT PORTAL
-- Database Engine Compatibility: PostgreSQL / MySQL / SQLite

-- 1. ADMINS TABLE (Dashboard Operators)
CREATE TABLE Admins (
    uid VARCHAR(128) PRIMARY KEY,
    email VARCHAR(255) NOT NULL UNIQUE,
    role VARCHAR(50) DEFAULT 'admin',
    createdAt TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- 2. EVENTS TABLE (Stores Event Specifications)
CREATE TABLE Events (
    id VARCHAR(50) PRIMARY KEY,
    title VARCHAR(255) NOT NULL,
    description TEXT,
    rules TEXT,
    prizes TEXT,
    teamSize INT DEFAULT 1,
    posterUrl LONGTEXT, -- Stores Base64 string of banners
    venue VARCHAR(255) NOT NULL,
    event_date DATE NOT NULL,
    event_time VARCHAR(50),
    registrationFee INT DEFAULT 0,
    lastRegistrationDate DATE,
    category VARCHAR(100) DEFAULT 'Hackathon',
    status VARCHAR(50) DEFAULT 'open' CHECK (status IN ('open', 'closed', 'completed')),
    seatsAvailable INT NOT NULL,
    totalSeats INT NOT NULL,
    createdAt TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

-- 3. COORDINATORS TABLE (1-to-Many Relationship with Events)
CREATE TABLE Coordinators (
    id INT AUTO_INCREMENT PRIMARY KEY,
    eventId VARCHAR(50) NOT NULL,
    name VARCHAR(255) NOT NULL,
    phone VARCHAR(20) NOT NULL,
    FOREIGN KEY (eventId) REFERENCES Events(id) ON DELETE CASCADE
);

-- 4. PARTICIPANTS TABLE (Stores Unique Attendee Profiles)
```

```
CREATE TABLE Participants (  
  email VARCHAR(255) PRIMARY KEY,  
  name VARCHAR(255) NOT NULL,  
  college VARCHAR(255) NOT NULL,  
  phone VARCHAR(20) NOT NULL,  
  updatedAt TIMESTAMP DEFAULT CURRENT_TIMESTAMP ON UPDATE CURRENT_TIMESTAMP  
);
```

-- 5. REGISTRATIONS TABLE (Transaction Receipts and Tickets)

```
CREATE TABLE Registrations (  
  id VARCHAR(50) PRIMARY KEY,  
  registrationId VARCHAR(50) NOT NULL UNIQUE, -- Unique Ticket ID  
  eventId VARCHAR(50) NOT NULL,  
  eventTitle VARCHAR(255) NOT NULL,  
  name VARCHAR(255) NOT NULL,  
  collegeName VARCHAR(255) NOT NULL,  
  department VARCHAR(255) NOT NULL,  
  year VARCHAR(50) NOT NULL,  
  email VARCHAR(255) NOT NULL,  
  phone VARCHAR(20) NOT NULL,  
  teamMembers TEXT,  
  paymentScreenshot LONGTEXT, -- Stores Base64 receipt proof  
  timestamp TIMESTAMP DEFAULT CURRENT_TIMESTAMP,  
  FOREIGN KEY (eventId) REFERENCES Events(id) ON DELETE CASCADE,  
  FOREIGN KEY (email) REFERENCES Participants(email) ON DELETE CASCADE  
);
```